

Parent Guide — senior / module-11- multiplication-near-base

IKS Curriculum

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Parent Guide — Multiplication Near a Power of 10 — Nikhilam (Senior)

What is your child learning?

When both factors are within a few units of a power of 10 (e.g. 97×96 , 103×104), Tirthaji's Nikhilam sūtra gives a two-step mental method that is much faster than long multiplication. Uses deficits/excesses from a base, with cross-subtraction on one side and product-of-deficits on the other.

Why this matters

This module is part of an Indian Knowledge Systems curriculum that introduces classical Indian frameworks alongside their modern-scientific parallels. Your child will learn to (a) name and use precise Sanskrit terms, (b) understand the observational basis for the classical framework, and (c) relate it to current scientific or scholarly understanding.

What you can do at home

Your child can now multiply numbers like 97×96 mentally. Test them at the shop or during meals. The goal is fluent number sense, not memorisation of a trick.

Conversation prompts

Try one of these over a meal or walk:

1. “What’s one Sanskrit word you learned this week? What does it mean?”
2. “What did you find surprising in this module?”
3. “Is there anything you’re not sure about? Let’s look it up together.”
4. “How would you explain this to your grandmother / grandfather?”

Questions parents often ask

Q: Is this religious instruction? A: No. The module presents Indian classical frameworks (cosmology, mathematics, astronomy, ecology) as historical knowledge systems with their own logic and method. The framing is scholarly and comparative, not devotional.

Q: Will my child be tested on Sanskrit? A: Sanskrit terms are introduced for precision, not for rote memorisation. The summative quiz asks students to *use* the terms — for example, “Which doṣa is dominant in the morning hours?” — rather than translate them.

Q: How does this connect to NCERT / board syllabus? A: This module supplements (does not replace) NCERT content. NEP 2020 §4.6 calls for the integration of Indian Knowledge Systems. The pack’s README.md lists the specific NEP principles each module advances.

Reading further (for parents)

See sources.md in this pack for the full source list.

10-Day Lesson Plan

Lesson Plan — Multiplication Near a Power of 10 — Nikhilam (Senior)

10-day arc, 45 min per session.

Day 1 — Finger-Trick Prelude

Objective: Practice the below-10 finger method (e.g., 7×8 using deficits).

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 2 — Two-Digit $\times$ Two-Digit Near 100 (Below)

Objective: Compute 97×96 using deficits.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 3 — Two-Digit $\times$ Two-Digit Near 100 (Above)

Objective: Compute 103×104 using excesses.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 4 — Mixed — One Above, One Below

Objective: Compute 102×98 with sign care.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 5 — 3-Digit $\times$ 3-Digit Near 1000

Objective: Compute 997×996 mentally.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 6 — Why It Works — The Algebra

Objective: Derive $(B-a)(B-b) = B(B-a-b) + ab$.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 7 — When To Use It

Objective: Identify problems where Nikhilam is faster than long multiplication.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 8 — Speed Sprints

Objective: Race traditional vs Nikhilam on 30 problems.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 9 — Mixed Practice + Module 7 Crossover

Objective: Solve problems combining $\times 11$ and near-base methods.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Day 10 — Final Assessment

Objective: Summative quiz and method-comparison reflection.

Flow: Warm-up (5 min) · Core (20 min) · Activity (15 min) · Wrap-up (5 min)

Pacing notes

- Senior sessions are 45 min. Adjust if your block is different.
- Days 5 and 9 are buffer days — use to reinforce weaker concepts or extend stronger ones.
- The summative project (see [assessments/project-brief.md](#)) runs in parallel from Day 6 onward.

Differentiation

- **For advanced students:** see [teacher-notes.md](#) for extension prompts.
- **For students needing more support:** simplify activities, do more pair-work, allow oral instead of written responses.

Homework

Homework — Multiplication Near a Power of 10 — Nikhilam (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — Finger-Trick Prelude** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Two-Digit $\times$ Two-Digit Near 100 (Below)** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Two-Digit $\times$ Two-Digit Near 100 (Above)** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Mixed — One Above, One Below** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — 3-Digit $\times$ 3-Digit Near 1000** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Why It Works — The Algebra** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — When To Use It** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Speed Sprints** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — Mixed Practice + Module 7 Crossover** · Write 2 sentences about today's concept. (~10 min)
10. **Day 10 — Final Assessment** · Write 2 sentences about today's concept. (~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.